

Computer Networks: Fundamentals of IP Addressing & Subnetting

Comprehensive Study Notes

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1 Network Layer & IP Addressing Basics

At the **Network Layer (Layer 3)** of the OSI model, data is encapsulated into a Protocol Data Unit (PDU) known as a **Packet**.

Every packet header contains essential control metadata, most notably the **Source IP Address** (where the packet originated) and the **Destination IP Address** (where the packet is headed).

1.1 Anatomy of an IPv4 Address

An IPv4 address is a **32-bit binary number** assigned to network interfaces. To make it human-readable, the 32 bits are divided into **4 equal groups** of 8 bits each, separated by dots (Dotted-Decimal Notation).

Key Term: Octet

Each 8-bit group in an IPv4 address is called an **Octet** because it consists of 8 binary digits ($32 \text{ bits} \div 4 = 8 \text{ bits/octet}$).

$\underbrace{11000000}_{\text{1st Octet (8 bits)}} . \underbrace{10101000}_{\text{2nd Octet (8 bits)}} . \underbrace{00001010}_{\text{3rd Octet (8 bits)}} . \underbrace{00001000}_{\text{4th Octet (8 bits)}}$

1.2 Binary to Decimal Conversion

Each bit position in an 8-bit octet represents a power of 2 (2^7 down to 2^0):

Bit Position	7	6	5	4	3	2	1	0
Positional Value	128	64	32	16	8	4	2	1

Example Conversion: Convert binary '11000000' to decimal:

$$(1 \times 128) + (1 \times 64) + (0 \times 32) + (0 \times 16) + (0 \times 8) + (0 \times 4) + (0 \times 2) + (0 \times 1) = 128 + 64 = \mathbf{192}$$

2 Network Portion vs. Host Portion

Every end-device (PC, server, smartphone, printer) and router interface connected to a network requires a unique IP address.

- **End Devices & Routers:** Require IP addresses at Layer 3.
- **Switches & Bridges (Layer 2):** Typically do not require an IP address to forward traffic (except for optional management interfaces).

An IP address is split logically into two structural parts:

1. **Network Portion (Net ID):** Identifies the specific physical or logical network. All devices on the same local network **must share the exact same network portion**.
2. **Host Portion (Host ID):** Identifies a specific individual device on that network. Every device on the same network **must have a unique host portion**.

2.1 Understanding Device Relationships in a Subnet

The table below illustrates how matching or differing Network and Host portions affect device communication:

Table 1: Comparison of Network and Host Bit Combinations

Network Bits	Host Bits	Relationship	Communication Result
Same	Different	Devices on Same Network	Devices can communicate directly at Layer 2 without a router.
Different	Same	Coincidental Host Match	Devices are on totally separate networks. Traffic must pass through a Router.
Different	Different	Distinct Devices & Nets	Normal internet setup; traffic must be routed across networks.
Same	Same	IP Address Conflict!	Invalid! Duplicate IP error; network communication fails for both devices.

3 Prefix Mask & Subnet Mask

How does a computer know where the Network portion ends and the Host portion begins? It uses the ****Subnet Mask**** or ****Prefix Length****.

3.1 Prefix Length (CIDR Notation)

Represented as a forward slash followed by a number (e.g., '/24'), the prefix length indicates the total number of consecutive bits reserved for the ****Network Portion**** starting from the left. * For ****192.168.10.8/24****: The '/24' means the first ****24 bits**** are Network bits, and the remaining ****8 bits**** (32 – 24) are Host bits.

3.2 Subnet Mask Representation

A Subnet Mask is a 32-bit number formed by setting all network bits to '1' and all host bits to '0'.

Example: Converting Prefix /24 to Subnet Mask

- **Prefix Length:** '/24' → Write twenty-four '1's followed by eight '0's.
- **Binary Form:** $\underbrace{11111111}_{255} . \underbrace{11111111}_{255} . \underbrace{11111111}_{255} . \underbrace{00000000}_0$
- **Dotted-Decimal Form:** 255.255.255.0

Reversing the Math: Finding Prefix Length from Subnet Mask

Given Subnet Mask: **255.255.240.0**

1. Convert to binary: 11111111.11111111.11110000.00000000
2. Count the contiguous '1's: $8 + 8 + 4 + 0 = 20$
3. **Prefix Length:** /20

4 Finding the Network IP: Bitwise AND Operation

To determine which network an IP address belongs to, devices perform a logical **bitwise AND** operation between the IP Address and the Subnet Mask.

Why do we use the AND Operation?

In Boolean logic, the truth table for an **AND** operation is:

$$1 \text{ AND } 1 = 1 \quad | \quad 1 \text{ AND } 0 = 0 \quad | \quad 0 \text{ AND } 0 = 0$$

Mathematical Principle:

- Any bit ANDed with '1' retains its original value ($x \text{ AND } 1 = x$).
- Any bit ANDed with '0' is forced to zero ($x \text{ AND } 0 = 0$).

Because a Subnet Mask consists of '1's for network bits and '0's for host bits, performing a bitwise AND **preserves the network portion exactly as-is** while **zeroing out all host bits**, yielding the pure Network Address!

4.1 Step-by-Step AND Operation Example

Given IP Address **192.168.10.35/24**:

IP Address (192.168.10.35):	11000000	10101000	00001010	00100011
Subnet Mask (255.255.255.0):	11111111	11111111	11111111	00000000
Resulting Network IP (AND):	11000000	10101000	00001010	00000000

Converting the result back to decimal yields: **192.168.10.0**

5 Types of IP Addresses in a Subnet

Within any given network block, addresses fall into three functional categories:

1. **Network Address:** The very first IP in the range. All host bits are set to '0'. Used strictly to identify the network. **Reserved—cannot be assigned to any device.**
2. **Broadcast Address:** The very last IP in the range. All host bits are set to '1'. Used to send messages to all hosts on the subnet. **Reserved—cannot be assigned to any device.**
3. **Usable Host Addresses:** All IP addresses residing strictly between the Network Address and Broadcast Address.

5.1 Formula for Calculating IP Capacity

To find the total number of IP addresses in a network given a prefix length P :

$$\text{Total IPs} = 2^{(32-P)}$$

$$\text{Usable Host IPs} = 2^{(32-P)} - 2$$

Note: We subtract 2 because the Network IP and Broadcast IP are reserved.

Mathematical Example

For a **192.168.1.0/26** subnet:

- Prefix length $P = 26$. Host bits $= 32 - 26 = 6$ bits.
- Total IPs $= 2^6 = 64$ total addresses.
- Usable Host IPs $= 64 - 2 = 62$ usable addresses.
- **Network Address:** '192.168.1.0' (Host bits all 0)
- **First Usable Host IP:** '192.168.1.1'
- **Last Usable Host IP:** '192.168.1.62'
- **Broadcast Address:** '192.168.1.63' (Host bits all 1)

6 Transmission Modes (Casting) & Special Addresses

6.1 Types of Message Delivery (Casting)

1. **Unicast (One-to-One):** A message sent from one single sender to one single recipient host.
2. **Broadcast (One-to-All):** A message sent from one host to every host in a network.
 - **Limited Broadcast:** Sent to all devices on the local network using Destination IP 255.255.255.255. Routers ****never**** forward limited broadcasts (non-routable).

- **Directed Broadcast:** Sent from a host on one network to all devices on a distant target network using that remote network's Broadcast IP (e.g., sending to 192.168.2.255).

3. Multicast (One-to-Many): A message sent from one host to a selected group of subscribed hosts. Uses dedicated **Class D** IP addresses ranging from **224.0.0.0** to **239.255.255.255**.

6.2 Loopback Address

* **Range:** 127.0.0.0/8 (most commonly 127.0.0.1). * **Purpose:** Internal testing, software troubleshooting, and local server communication on the host machine. * **Key Characteristic:** Traffic sent to a loopback address **never** leaves the end device's local network interface card (NIC).

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7 Anatomy & Rules of IPv6 Addresses

Due to IPv4 address exhaustion, **IPv6** was developed using a **128-bit address space** (providing 2^{128} or $\approx 3.4 \times 10^{38}$ unique addresses).

7.1 IPv6 Structure

* Total length: **128 bits**. * Divided into **8 chunks** (often called **hextets** or **hexstates**) of **16 bits each**. * Written in **Hexadecimal** format (where 4 binary bits = 1 Hex digit → 4 Hex digits per chunk). * Chunks are separated by colons (:).

Example Uncompressed IPv6 Address: **2001 : 0db8 : 0000 : 0000 : 0000 : 0000 : 1428 : 57ab**

7.2 IPv6 Compression Rules

To make long IPv6 addresses easier to read and write, two official compression rules are applied:

Rule 1: Omit Leading Zeros

Any leading zeros in any 16-bit hex segment can be omitted.

- 0db8 → db8
- 0000 → 0
- 00ab → ab

Note: Trailing zeros cannot be omitted!

Rule 2: Double Colon (::) Substitution

A single contiguous string of two or more zero-value hex segments (0000) can be replaced with a double colon (::).

CRITICAL RULE: The double colon (::) can **ONLY** be used **ONCE** per IPv6 address.

Why? Using it more than once creates ambiguity during decompression, making it mathematically impossible to determine how many zero bits belong to each section.

7.3 Step-by-Step Compression Example

Compressing an IPv6 Address

Original Address:

2001 : 0db8 : 0000 : 0000 : 000f : 0000 : 0000 : 0001

1. Apply Rule 1 (Drop leading zeros):

2001 : db8 : 0 : 0 : f : 0 : 0 : 1

2. Apply Rule 2 (Use :: on the largest contiguous group of zeros): Notice we have two groups of zeros: two contiguous zeros (0:0) and another two contiguous zeros (0:0). We pick the first group to replace with ::

2001 : db8 :: f : 0 : 0 : 1